Qinzhi Peng

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EDUCATION

University of California, Santa Cruz Santa Cruz, CA
Ph.D. of Statistics September 2025 - June 2030

Duke University Durham, NC
Master of Statistical Science August 2023 - May 2025

- **Major GPA:** 4.00/4.00
- **Relevant Courses:** Bayesian Statistics, Multilevel and Hierarchical Models, High-dimensional Statistics

Carnegie Mellon University Pittsburgh, PA
Master of Information Systems Management August 2021 - December 2022

- **Major GPA:** 3.60/4.00
- **Relevant Courses:** Machine Learning, Convex & non-convex Optimization, Data Mining

Beijing University of Technology Beijing, China
Bachelor of Science in Mathematics and Applied Mathematics September 2017 - July 2021

- **Major GPA:** 3.83/4.00
- **Relevant Courses:** Real Analysis, Probability Theory, Mathematical Statistics, Regression Analysis
- **Awards:** 2017 – 2021 Annual Academic Scholarship (Top 5%)

RESEARCH EXPERIENCE

Bayesian Inference with Optimization Durham, NC
Research Assistant | Duke University February 2024 - December 2024

- Explored the application of a bridged posterior framework that integrates probabilistic modeling with an optimization sub-problem.
- Designed a Bayesian semi-supervised logistic classifier, and used the relax-and-split approach to optimize parameters of interest for real-world datasets with sparse labeled data.
- Improved 0.5% classification accuracy compared to the Bayesian maximum margin classifier and other traditional methods through proper prior and hyperparameter setting
- Conducted a simulation study using a Bayesian generalized linear mixed-effects model to compare two predictive methods based on MCMC results.

Random Projection Forest in high-dimensional space Beijing, China
Research Assistant | Renmin University March 2023 - August 2023

- Developed ensemble classifiers combining decision trees trained on random subspaces for high-dimensional classification problems.
- Designed and implemented sparse random matrices to project high-dimensional feature vectors into lower-dimensional spaces.
- Achieved an average of 1% improvement in classification accuracy compared to traditional classifiers using UCI Machine Learning Repository datasets.
- Analyzed the impact of different distributed random projections and demonstrated the consistency of random projection forests.

Generative Face Completion Beijing, China
Research Assistant | China Academy of Electronics and Information Technology June 2020 - August 2020

- Utilized PyTorch to improve the network structure in the generative face completion algorithm.
- Optimized the face completion algorithm by adding landmark loss, resulting in 0.5% and 0.8% improvement in Peak signal-to-noise ratio and Structural Similarity, respectively.

Application of big data technology in basketball games Beijing,
China
Research Assistant | Beijing University of Technology June 2019 - November 2019

- Analyzed the application of big data technology in basketball games.
- Proposed a University Player Efficiency Rating model and created a heatmap based on Field Goal Percentages.
- Made recommendations to the school's basketball team based on the results.

PROFESSIONAL EXPERIENCE

Global AI Corporation New York, NY
Data Scientist Intern July 2022 - August 2022

- Conducted a comprehensive statistical analysis on time series data for the MSCI US index stocks from 2013-2020.
- Completed a detailed analysis of the influence of different risk types on the valuation and price returns of stocks, utilizing the SDGs as a risk factor framework.

China Unicom Big Data Co., Ltd. Beijing, China
Data Scientist Intern December 2020 - February 2021

- Regularly managed the spatiotemporal data of tourists from 35 million users using GeoMesa.
- Performed an assessment of features for tourists using the DBSCAN clustering algorithm to identify patterns and grouping.

TEACHING EXPERIENCE

STAT 5 Statistics | UCSC January 2026 - March 2026
STAT 5 Statistics | UCSC September 2025 - December 2025
STA 332 Statistical Inference | Duke University August 2024 - December 2024
STA 532 Theory of Statistical Inference | Duke University August 2024 - December 2024

PROJECTS

Analysis of Breeding Success in Song Sparrows, Duke October 2024 - December 2024

- Analyzed a 15-year observational dataset on song sparrows to identify key factors affecting reproductive success.
- Built a nested hierarchical model, including fixed effects and random effects, to account for year-to-year variability and individual-level covariates.

Stock Price Forecasting with Multi-Order HMMs, Duke, Team of 3 October 2024 - December 2024

- Developed first-order and high-order Hidden Markov Models to forecast the stock price of NVIDIA in 5 years.
- Evaluated model performance using Mean Absolute Percentage Error (MAPE) and Directional Accuracy (DA), achieving the lowest MAPE of 2.02% and highest DA of 61.71 % with the second-order model.

Capstone Project with Motif Research, Inc., DBA Artifact, CMU, Team of 5 September 2022 - December 2022

- Developed an automatic text classifier based on additive logistic regression to identify valuable texts.
- Achieved a 1.2% improvement in the identification of useful texts for 100k new texts compared to the current model.

TECHNICAL SKILLS

Programming Languages: R, Python, MATLAB, SQL, Java, JavaScript
Technologies: PyTorch, Scikit-learn, Matplotlib
Data Management & Analytics: Oracle, MySQL, MongoDB, BigQuery, Apache Spark